

Test Plan

The suite has **154 tests** across 16 files. They are the executable evidence that the honesty and capability-tier guarantees in this documentation actually hold.

Running

```
cd airone
```

```
AIRONE_JWT_SECRET=test_secret_key_that_is_definitely_long_enough_123456 \  
QT_QPA_PLATFORM=offscreen python -m pytest -q
```

CI runs the same suite on Python 3.10/3.11/3.12 (see [.github/workflows/ci.yml](#)).

Coverage by file

File	Tests	What it proves
test_protocol.py	7	Frame build/parse round-trips, CRC detection, resync, duplicate/out-of-order flagging.
test_measurements.py	5	Measurement/QualityState coherence — invalid data can never masquerade as valid.
test_pipeline.py	5	The 20-stage pipeline tags quality correctly and isolates per-stage failures.
test_mission_state_machine.py	5	Only legal mission-state transitions are allowed.
test_security.py	7	bcrypt hashing, JWT create/decode/expiry, RBAC gating, audit hash chain, secret enforcement.
test_scientific.py	10	Representative analyses compute correct values with propagated uncertainty and honest insufficiency.
test_scientific_worker_integraction.py	2	End-to-end Tier-1→Tier-2: simulated packets → real pipeline → analyses → persistence.
test_simulation.py	7	ISA atmosphere, sensor models, and flight profile behave physically; outputs are labelled SIMULATED.
test_ml.py	17	Detector correctness, no-imputation feature extraction, checksum tamper refusal, honest PSI drift, explicit training failures, and the advisory-only

File	Tests	What it proves
		invariant.
test_api.py	11	Response envelope, 401/403/404/422/503 failure modes, and NOT_CONFIGURED honesty at the HTTP boundary.
test_gui_smoke.py	4	The GUI builds all tabs headlessly and degrades honestly when the API is unreachable.

Key invariants asserted

- **Advisory-only ML:** test_ml.py asserts the ML worker holds no mission-machine handle and that no src/ml/ file references the mission state machine at all.
- **No imputation:** feature extraction drops incomplete/invalid frames and counts them; it never fabricates a value.
- **Integrity:** a tampered model artefact fails to load (SafeLoadError).
- **Honest drift:** PSI returns None (→ UNAVAILABLE) on insufficient data, never a fabricated 0; identical distributions classify STABLE, shifted ones SIGNIFICANT.
- **Honest API:** training with no data returns 422 with a precise reason; a missing worker returns 503 NOT_CONFIGURED — never a fake success.

Security hardening tests

- test_security_hardening.py — password policy (length, classes, deny-list, username), UserStore CRUD/persistence/dev-account gate, LoginThrottle back-off and per-IP lock, JWT iss/aud/type/jti/algorithm pinning (none, HS512 rejected), lifetime caps, persistent revocation and pruning.
- test_api_security.py — unknown-field rejection, body cap (413), CORS wildcard refusal, security headers, X-Request-ID sanitisation, generic login errors, lockout (429 + Retry-After), disabled accounts, refresh rotation and reuse detection, must-change flow, admin user management, immutable/redacted config, honest /status.
- test_telemetry_hardening.py — header pre-check counters, replay flag, HMAC accept/reject/UNVERIFIABLE/required, hostile JSON payloads (size, depth, type, provenance upgrade, timestamps), AuthenticateStage verdicts, authenticated simulator frames, loopback bind resolution, audit-chain verification (tamper, deletion, restart, CLI), launcher bootstrap CLI.
- test_firmware_frame_parity.py — C/Python frame parity including the authenticated frame, SHA-256 and RFC 4231 HMAC vectors.

Expected result

154 passed

Any failure should be treated as a real regression, not flaked away.